

Fairness Intervention Workflow

Purpose

This document defines the operational flow governing how fairness issues are evaluated, addressed, validated, and approved.

It establishes:

- Sequencing logic
- Decision gates
- Escalation conditions
- Iteration rules

Detailed methodologies and documentation requirements are defined in their respective stage documents.

Workflow Structure

The lifecycle follows a controlled progression:

Disparity Identified → Causal Diagnosis → Risk Classification → Intervention Selection → Implementation → Validation → Governance Review → Monitoring

The workflow is decision-driven and risk-proportionate.

End-to-End Fairness Workflow

A[Disparity Detected] --> B[Causal Fairness Analysis]

B --> C[Risk Classification]

C --> D[Intervention Selection]

D --> E{Intervention Type}

E -->|Data Bias| F[Pre-Processing]

E -->|Model Bias| G[In-Processing]

E -->|Decision Bias| H[Post-Processing]

E -->|Multi-Stage Bias| I[Layered Intervention]

F --> J[Implementation]

G --> J

H --> J

I --> J

J --> K[Validation]

K --> L{Validation Passed?}

L -->|No| B

L -->|Yes| M[Governance Approval]

M --> N[Deployment & Monitoring]

N -->|Drift Detected| B

Cross-Toolkit Workflow Mapping

Step	Action	Primary Toolkit	Output
Disparity Detection	Identify fairness issue	Scrum + Monitoring	Bias signal
Causal Diagnosis	Identify root cause	Architecture	Causal model
Risk Classification	Assign system risk	Governance	Risk level
Intervention Selection	Choose strategy	Architecture	Intervention plan
Implementation	Apply interventions	Engineering + Scrum	Updated system
Validation	Evaluate fairness & performance	Validation Framework	Pass/Fail decision
Governance Approval	Review and authorize	Compliance	Release approval
Monitoring	Track drift and fairness	Governance	Alerts & triggers

Step 1 — Disparity Identification

The workflow begins when:

- A fairness disparity is detected, or
- Monitoring signals deviation, or
- A formal review is initiated.

If no material disparity is confirmed, monitoring continues without intervention.

Step 2 — Causal Diagnosis

A structured causal analysis must determine:

- Affected groups

- Root cause mechanisms
- The stage(s) at which bias enters

No intervention may proceed without a causal diagnosis.

(See: Causal Fairness Analysis)

Step 3 — Risk Classification

Before selecting an intervention strategy, the system must be classified according to risk level.

Risk classification determines:

- Required evaluation scope
- Governance intensity
- Whether stage-skipping is permissible

(See: Risk-Based Intervention Selection)

Step 4 — Intervention Pathway Determination

Intervention pathways are selected based on:

- Identified bias stage(s)
- System risk classification
- Technical constraints

Possible pathways:

- Single-stage intervention
- Layered multi-stage intervention
- Constrained intervention (e.g., black-box system)

Stage selection must align with causal findings.

This workflow does not assume all stages are always required.

Step 5 — Intervention Execution

Selected intervention stage(s) are implemented according to their respective documents:

- Pre-Processing
- In-Processing
- Post-Processing

If a stage is not selected:

- Justification must be documented
- Risk-based constraints must be respected

Execution details are defined in stage-specific documents.

Step 6 — Validation Gate

Before deployment:

- Fairness outcomes must be evaluated
- Performance trade-offs must be assessed
- Stability must be confirmed

If validation fails, the workflow returns to Causal Diagnosis.

This workflow is iterative.

(See: Validation & Monitoring)

Step 7 — Governance Approval

Deployment requires:

- Completed documentation
- Recorded intervention rationale
- Required approvals based on system risk

High-Risk systems require elevated governance review.

Deployment is prohibited without formal approval.

(See: Final Intervention Report)

Step 8 — Post-Deployment Monitoring

After release:

- Fairness must be continuously monitored
- Drift must be tracked
- Renewed disparities trigger re-entry into this workflow

Fairness oversight is ongoing.

Escalation Conditions

Escalation is mandatory when:

- Validation results are inconclusive
- Trade-offs exceed acceptable thresholds
- Bias persists after intervention
- Required stages are proposed to be skipped in High-Risk systems

Escalation authority is defined in governance documents

Iteration Principle

At any stage, if:

- Fairness objectives are not met, or
- New bias is detected,

The system must return to Causal Diagnosis.

Intervention is not a one-time event.